

MOHAMED MOSTAFA

Software Engineer

Banha, Al Qalyubia, Egypt | mohamed.mostafa0699@gmail.com | +20 114 231 4446

[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

Software Engineer with 4+ years designing, building, and operating production systems end-to-end — API and data architecture, real-time infrastructure, payment processing, and cloud deployment. Sole engineer for two live marketplace platforms handling 30K-120K requests/day, holding full ownership of architecture and tooling decisions and independently resolving production incidents. Builds across the stack in independent work — Go network services, CRDT-based real-time collaboration systems, and open-source developer tooling used outside the day job.

EXPERIENCE

Software Engineer | Lesoll

Jun 2023 – Jul 2026 *Cairo, Egypt*

Sole backend engineer for two live production platforms (Lesoll marketplace, EGYStay hospitality/property management), with ownership of backend architecture, infrastructure, and tooling decisions.

- Designed and rebuilt backend architecture from the ground up using Node.js, Express.js, TypeScript, and MongoDB, including REST APIs, authentication, authorization, and database design.
- Owned backend architecture for two production platforms serving 30K-120K requests/day; cut key-endpoint latency from 680ms/1.2s to 234ms/138ms and reduced database load by 40% through Redis caching, query optimization, and BullMQ background processing.
- Independently diagnosed and resolved two critical production incidents with no external escalation: a runaway N+1 query that took the database fully offline, and an image-handling defect affecting live listings.
- Built real-time features with Socket.IO — chat, notifications, and live user presence — across both platforms.
- Integrated PayMob and Amazon Payment Services, building secure webhook processing with HMAC verification and idempotency ledgers to ensure idempotent payment processing and prevent duplicate transaction handling.
- Owned production infrastructure on AWS (EC2, S3, CloudFront, Route 53) with Docker, Nginx, PM2, and GitHub Actions CI/CD; built automated MongoDB backups to S3 with SSE-KMS encryption.
- Improved organic discoverability by optimizing API response performance and generating dynamic XML sitemaps for active listings.

PROJECTS

SRVJ — Live, in active development srvj.elrefai.me

- Built real-time collaborative diagram editing using Yjs CRDTs over a custom WebSocket server implementing the Yjs sync and awareness protocols, enabling live cursors, presence, and multiplayer editing.
- Secured connections with PASETO v4 authentication and project-level role-based access control, isolating collaboration sessions into dedicated boardrooms.
- Designed a dual-snapshot persistence layer storing both the authoritative Yjs binary state for lossless recovery and a denormalized JSON view for efficient querying.
- Architected the backend with PostgreSQL/Prisma for relational data (users, projects, memberships) and MongoDB/Mongoose for high-frequency diagram mutations.
- Built horizontally scalable notifications (BullMQ workers → Redis Pub/Sub → SSE fan-out) and transactional email templates with cross-client (MSO/Outlook) compatibility.
- Hardened with Zod validation, Helmet, CORS, rate limiting, centralized error handling, Pino structured logging, and Prometheus metrics; containerized with Docker Compose behind Nginx.

Leoxy — [GitHub](#)

- Built a production-grade HTTP reverse proxy in Go with path-prefix-based routing across configurable upstream servers and a Cobra-based CLI (run/start/stop as a managed background process).
- Implemented a security layer supporting per-IP and global rate limiting (token bucket, Redis-backed for multi-instance deployments), API-key and JWT (HS256) authentication, and optional mTLS.
- Added health-check and liveness endpoints, structured request logging, and explicit read/write/idle timeouts for production safety.

Data-Localizer — [GitHub](#)

- Published an open-source TypeScript package for locale-aware data handling in Node.js applications; currently developing a new major version with architectural changes. [Confirm exact scope/description before sending — see note]

Gen-Import — Open source, CLI, npm [GitHub](#)

- Built an open-source CLI that automatically generates index.ts barrel exports for large TypeScript codebases via recursive directory traversal with configurable exclusion rules, cutting manual maintenance and import errors.
- Added file-watching for live regeneration and packaged it as a zero-runtime, drop-in tool for existing Node.js/TypeScript workflows.

EDUCATION

Bachelor of Computer Science — Modern Academy, Maadi, Cairo · Jul 2022

Coursework: Databases, Operating Systems, OOP, Data Structures, Distributed Systems

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Go, Python, C++

Backend: Gin(Go), Node.js, Express.js, REST API design, WebSockets (Socket.IO), SSE

Real-Time & Distributed Systems: CRDTs (Yjs), Redis Pub/Sub fan-out, event-driven architecture, idempotency patterns

Databases & Caching: MongoDB, PostgreSQL, Prisma, Mongoose, Redis

Messaging & Async Processing: BullMQ, AWS SQS, Redis Pub/Sub

Payments: PayMob, Amazon Payment Services, Stripe (webhooks, HMAC, idempotency)

Cloud & DevOps: AWS (EC2, EKS, S3, ECR, Lambda, CloudFront, Route 53), Docker & Docker Compose, GitHub Actions CI/CD, Nginx, PM2

Observability: Prometheus, Pino/structured logging

Frontend (independent projects): Vue.js, Vite, TailwindCSS, UnoCSS